$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$e^{i\pi} + 1 = 0$$

$$e^{ix} = \cos x + i\sin x$$

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k$$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$$

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$$

$$\lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n = e$$

$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$\frac{d}{dx}\left(x^n\right) = nx^{n-1}$$

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$$

$$\int u\,dv = uv - \int v\,du$$

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

$$\frac{\partial u}{\partial t} = \alpha \nabla^2 u$$

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_p \frac{1}{1-p^{-s}}$$

$$\Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t}\,dt$$

$$n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$$

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x \xi}\,dx$$

$$f(a) = \frac{1}{2\pi i} \oint_\gamma \frac{f(z)}{z-a}\,dz$$

$$\left|\sum_{i=1}^{n} a_i b_i\right|^2 \le \sum_{i=1}^{n} a_i^2 \sum_{i=1}^{n} b_i^2$$

$$|x+y| \le |x| + |y|$$

$$\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p}$$

$$\sqrt[n]{x_1 x_2 \cdots x_n} \le \frac{x_1 + x_2 + \cdots + x_n}{n}$$

$$a^2 + b^2 = c^2$$

$$\sin^2\theta + \cos^2\theta = 1$$

$$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$$

$$\tan\theta = \frac{\sin\theta}{\cos\theta}$$

$$c^2 = a^2 + b^2 - 2ab\cos\gamma$$

$$\log_b x = \frac{\ln x}{\ln b}$$

$$\sqrt{2} = 1 + \cfrac{1}{2 + \cfrac{1}{2 + \cdots}}$$

$$\phi = \frac{1 + \sqrt{5}}{2}$$

$$F_n = \frac{\phi^n - \psi^n}{\sqrt{5}}$$

$$\det(A - \lambda I) = 0$$

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

$$R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$$I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$A\mathbf{x} = \lambda\mathbf{x}$$

$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos\theta$$

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

$$\delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

$$f(n) = \begin{cases} n/2 & \text{if } n \text{ is even} \\ 3n + 1 & \text{if } n \text{ is odd} \end{cases}$$

$$\forall \varepsilon > 0 \ \exists \delta > 0 : |x - a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$\overline{A \cup B} = \overline{A} \cap \overline{B}$$

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$$

$$f : \mathbb{R}^n \to \mathbb{R}^m$$

$$x \in [a, b] \iff a \leq x \leq b$$

$$p \Rightarrow q \equiv \neg p \lor q$$

$$\neg(p \land q) \equiv \neg p \lor \neg q$$

$$a \equiv b \pmod{n}$$

$$\gcd(a, b) = \gcd(b, a \bmod b)$$

$$a^{p-1} \equiv 1 \pmod{p}$$

$$\varphi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right)$$

$$\pi(x) \sim \frac{x}{\ln x}$$

$$P(A \mid B) = \frac{P(B \mid A) P(A)}{P(B)}$$

$$\mathbb{E}[X] = \sum_i x_i p_i$$

$$\operatorname{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$$

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2 \sum_i (y_i - \bar{y})^2}}$$

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

$$H(X) = -\sum_{x} p(x) \log p(x)$$

$$D_{\mathrm{KL}}(P \| Q) = \sum_{x} P(x) \log \frac{P(x)}{Q(x)}$$

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\operatorname{softmax}(z)_i = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

$$\theta_{t+1} = \theta_t - \eta \nabla_\theta L(\theta_t)$$

$$L = -\frac{1}{N} \sum_{i=1}^{N} \left[y_i \log \hat{y}_i + (1 - y_i) \log (1 - \hat{y}_i) \right]$$

$$\hat{\theta} = \arg\max_{\theta} \prod_{i=1}^{n} p(x_i \mid \theta)$$

$$\chi^2 = \sum_i \frac{(O_i - E_i)^2}{E_i}$$

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

$$E = mc^2$$

$$F = ma$$

$$F = G\frac{m_1 m_2}{r^2}$$

$$E = h\nu$$

$$\lambda = \frac{h}{p}$$

$$\Delta x \, \Delta p \geq \frac{\hbar}{2}$$

$$i\hbar\frac{\partial}{\partial t}\Psi = \hat{H}\Psi$$

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V\psi = E\psi$$

$$E_n = -\frac{13.6\,\text{eV}}{n^2}$$

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$$

$$E^2 = (pc)^2 + (mc^2)^2$$

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{B} = \mu_0\mathbf{J} + \mu_0\varepsilon_0\frac{\partial \mathbf{E}}{\partial t}$$

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}$$

$$ds^2 = -c^2dt^2 + dx^2 + dy^2 + dz^2$$

$$\Gamma^\lambda_{\mu\nu} = \frac{1}{2}g^{\lambda\sigma}\left(\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu}\right)$$

$$S = k_B \ln \Omega$$

$$PV = nRT$$

$$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$$

$$dU = T\,dS - P\,dV$$

$$\Delta G = \Delta H - T\Delta S$$

$$\langle E \rangle = \frac{3}{2}k_BT$$

$$Z = \sum_i e^{-\beta E_i}$$